

E-Waste Case Studies

MSE 2 Preparation

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EWEM

10 case studies | one slide each | report-format pointers

1 Why Japan Recycles More Than Most Countries: Lessons for India

EXECUTIVE SUMMARY

- Japan pairs strict household segregation, law-backed take-back and strong social norms to keep resources circulating.
- Result: one of the world's most mature systems, from neighbourhood sorting to the Tokyo 2020 e-waste medals.
- Lesson for India: rules work when consumers, producers and municipalities each have an enforced role.

BACKGROUND

- Small land area, few natural resources and landfill shortage pushed Japan to treat waste as a resource.
- Basic Act for a Sound Material-Cycle Society (2000) set the 3R framework: Reduce, Reuse, Recycle.
- Home Appliance Recycling Law (2001) and Small Home Appliance Recycling Law (2013) cover e-waste.

STRATEGY / INITIATIVE

- Households sort waste into many categories on fixed collection days; local governments enforce.
- Consumers pay a recycling fee when discarding TVs, ACs, fridges and washers; makers must take back and recycle.
- 'Urban mine' approach: Tokyo 2020 medals made from donated phones and small devices.

IMPACT & RESULTS

Indicator	Data / Outcome
Legal framework	3R Basic Act (2000); Appliance Recycling Law (2001)
Regulated appliances	AC, TV, fridge/freezer, washer/dryer
Tokyo 2020 medals	~5,000 medals from ~79,000 t of devices (~6.2 M phones)
Appliance recycling rates	Typically ~70-90%+ by weight (govt reports)

SWOT ANALYSIS

Strengths: Clear legal duties, high citizen discipline, dense collection and processing network

Weaknesses: Heavy sorting burden; disposal fee can encourage illegal dumping; high operating cost

Opportunities: Urban mining of metals; export of know-how; digital product tracking

Threats: Ageing workforce, rising small-device volumes, informal export of used appliances

COMPARISON WITH INDIA

- India: mostly mixed waste at source; SWM Rules 2016 segregation is weakly enforced.
- Japan: consumer pays at disposal. India: producers pay via EPR, consumers see no cost signal.
- India's informal sector handles most e-waste; Japan's chain is largely formal.

RECOMMENDATIONS

- Enforce source segregation with municipal penalties and incentives.
- Pilot deposit-refund or fee-at-disposal schemes in metros.
- Build formal drop-off networks and use informal collectors as partners.

CONCLUSION

- Japan shows recycling succeeds when law, infrastructure and culture work together.
- India can adapt segregation, take-back and awareness, scaled to local realities.

REFERENCES

- Ministry of the Environment, Japan - Home Appliance Recycling Law reports
- Tokyo 2020 Organising Committee - Medal Project
- UN Global E-waste Monitor (UNITAR/ITU)

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- Laws: 3R Basic Act 2000, HARL 2001, SHARL 2013.
- Model = consumer fee + manufacturer take-back + strict segregation.
- Tokyo 2020 medals = urban mining of e-waste.
- Lesson for India: enforcement + behaviour change.
- Likely question: compare Japan's and India's e-waste models.

2 Switzerland's E-Waste Success Story

EXECUTIVE SUMMARY

- Switzerland built one of the world's first national e-waste take-back systems, funded by consumers and run by industry.
- An advance recycling fee (ARF) pays for collection and treatment, so the system is self-financing.
- Per-capita collection is among the highest worldwide.

BACKGROUND

- SENS (fridges, from 1990s), SWICO (IT/office) and SLRS (lighting) began as industry schemes.
- The ORDEE ordinance (1998) made take-back legally mandatory.
- Wealthy, compact country with high electronics use and strong environmental awareness.

STRATEGY / INITIATIVE

- ARF is added to the price at purchase of each device.
- Retailers, manufacturers and importers must take back old devices free of charge.
- Producer organisations run logistics; certified recyclers follow strict standards.

IMPACT & RESULTS

Indicator	Data / Outcome
Model	Producer responsibility + Advance Recycling Fee
Legal basis	ORDEE ordinance (1998)
Collection	~15 kg per person per year (approx.)
Treatment	Certified recyclers; hazardous parts safely treated

SWOT ANALYSIS

Strengths: Self-financing ARF, free take-back, high public trust and compliance

Weaknesses: Relies on producer honesty; high labour and treatment costs

Opportunities: Better online-sale tracking; recovery of critical raw materials

Threats: Free-riders, cross-border online purchases, growing small-device volumes

COMPARISON WITH INDIA

- India: no consumer-visible fee; costs move through EPR certificate trading.
- Swiss retailers must accept old devices; Indian retail take-back is patchy.
- Swiss chain is formal end-to-end; India's is mostly informal.

RECOMMENDATIONS

- Pilot an advance recycling fee or deposit on selected products.
- Make retailer take-back mandatory and audited.
- Create a central body to audit recyclers and track material flows.

CONCLUSION

- Switzerland proves a producer-run, consumer-funded model can last for decades.
- India can borrow fee-plus-take-back logic while including its informal sector.

REFERENCES

- SWICO Recycling and SENS eRecycling - annual reports
- Swiss Federal Office for the Environment (FOEN) - ORDEE
- UN Global E-waste Monitor (UNITAR/ITU)

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- ORDEE 1998 made take-back mandatory.
- Three PROs: SENS, SWICO, SLRS.
- ARF = fee paid at purchase that funds recycling.
- Free take-back at retailers.
- Likely question: explain ARF and why it works.

3 India's E-Waste (Management) Rules 2022

EXECUTIVE SUMMARY

- The 2022 Rules replaced the 2016 Rules and made EPR certificate-based and digitally tracked.
- They apply from 1 April 2023 and widen scope to more products and more actors, including refurbishers.

BACKGROUND

- India generated ~1.6 million tonnes of e-waste in 2021-22 (CPCB).
- Earlier rules (2011, 2016) had weak enforcement and low formal collection.
- Notified by MoEFCC in November 2022.

STRATEGY / INITIATIVE

- Producers, manufacturers, refurbishers and recyclers register on the CPCB portal.
- EPR targets rise yearly: 60%, 70%, then 80% from 2025-26.
- Covers 106 items; CPCB sets certificate floor/ceiling prices; environmental compensation for violations.

IMPACT & RESULTS

Indicator	Data / Outcome
Items covered	21 (2016 Rules) to 106 (2022 Rules)
Effective date	1 April 2023
EPR targets	60% (2023-24), 70% (2024-25), 80% (2025-26 on)
E-waste generated	~1.6 Mt (2021-22, CPCB)

SWOT ANALYSIS

Strengths: Wider scope, digital tracking, certificate trade gives recyclers revenue

Weaknesses: Informal sector not fully integrated; complex compliance for small players

Opportunities: Growth of formal recycling, refurbishing and urban mining

Threats: Fake certificates, weak monitoring, informal leakage

COMPARISON WITH INDIA

- vs India's 2016 Rules: 106 items vs 21; certificate EPR vs collection-target EPR.
- vs EU WEEE: India adds price-controlled certificates but has weaker take-back infrastructure.
- New formal role for refurbishers supports reuse.

RECOMMENDATIONS

- Audit recyclers physically, not only by paperwork.
- Bring informal collectors in through registration and training.
- Run consumer campaigns on where and how to return devices.

CONCLUSION

- The 2022 Rules are a major upgrade on paper.
- Results depend on enforcement, traceability and household collection.

REFERENCES

- MoEFCC - E-Waste (Management) Rules, 2022
- CPCB - EPR portal and guidelines for e-waste
- UN Global E-waste Monitor (UNITAR/ITU)

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- Effective 1 April 2023; 106 items.
- Digital EPR certificates on CPCB portal.
- Actors: producer, manufacturer, refurbisher, recycler.
- Targets 60/70/80%.
- Likely question: critically evaluate the 2022 Rules.

4 India's Extended Producer Responsibility (EPR): Success or Paper Compliance?

EXECUTIVE SUMMARY

- EPR makes producers responsible for end-of-life products, but in India compliance is often on paper.
- Certificates can be bought without proof of physical recycling, so real recovery lags targets.

BACKGROUND

- EPR began in the 2011 Rules, was strengthened in 2016 and became certificate-based in 2022.
- Producers meet targets by buying certificates from registered recyclers.
- Informal sector still handles most e-waste.

STRATEGY / INITIATIVE

- CPCB portal registers producers and recyclers and issues EPR certificates.
- Recyclers generate certificates for processed e-waste; producers buy them to meet targets.
- Environmental compensation for shortfall or false reporting.

IMPACT & RESULTS

Indicator	Data / Outcome
Formal share of e-waste	Small; informal sector handles most (commonly cited >90%)
Recycler capacity	Authorised capacity far above actual throughput (reported)
Certificate integrity	Suspected fake or duplicate certificates flagged
Target design	Paper targets 60-80%; actual collection much lower

SWOT ANALYSIS

Strengths: Legal backing, digital portal, clear producer liability

Weaknesses: Paper compliance, weak audits, no push to reach households

Opportunities: QR/blockchain tracking, third-party audits, informal linkage

Threats: Fake certificates, price undercutting, informal channels persist

COMPARISON WITH INDIA

- Japan and Switzerland fund consumer-facing collection; India funds certificates.
- Global best practice audits material flows; India relies heavily on documents.
- India's EPR is recycler-centred, not collection-centred.

RECOMMENDATIONS

- Verify certificates with mass-balance data and site audits.
- Fund collection networks and informal aggregators directly.
- Publish recycler-wise data publicly.

CONCLUSION

- India's EPR is a promising framework but currently closer to paper compliance than proven success.
- Real success needs verified tonnage and household collection.

REFERENCES

- CPCB - EPR guidelines and portal
- Centre for Science and Environment (CSE) - EPR analyses
- MoEFCC - E-Waste Rules 2016 and 2022

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- EPR = producer pays for end-of-life management.
- Paper compliance = certificates without real recycling.
- Fixes: audits, traceability, fund collection.
- Contrast with Swiss ARF model.
- Likely question: is EPR in India a success?

5 Bengaluru: India's E-Waste Capital?

EXECUTIVE SUMMARY

- Bengaluru, India's IT capital, is a major urban e-waste generator and home to key formal recyclers.
- Corporate, household and start-up streams make it a testbed for formalisation.

BACKGROUND

- Large IT/BPO workforce, frequent device refresh and heavy corporate e-waste.
- KSPCB authorises recyclers; E-Parisaraa (Peenya) is among India's early authorised recyclers.
- Informal scrap markets coexist with formal facilities.

STRATEGY / INITIATIVE

- KSPCB-licensed recyclers and dismantlers process the formal stream.
- Corporate take-back and IT asset disposition with data destruction.
- NGOs and civic bodies run awareness and collection drives.

IMPACT & RESULTS

Indicator	Data / Outcome
State position	Karnataka among top e-waste generating states (CPCB)
Key areas	Peenya industrial area; Whitefield, Electronic City IT corridors
Formal recycler	E-Parisaraa Pvt Ltd (Peenya), KSPCB-authorised
Corporate stream	Large ITAD volumes with certified data destruction

SWOT ANALYSIS

Strengths: Concentrated IT waste, skilled formal recyclers, active NGOs

Weaknesses: Household e-waste still leaks to scrap dealers; low awareness

Opportunities: Corporate CSR tie-ups, drop-off points, urban mining

Threats: Fast device growth, informal dismantling, land and cost pressure

COMPARISON WITH INDIA

- Ahead of most Indian cities in formal recycling, behind global cities like Zurich or Tokyo.
- Corporate streams are formal; household streams remain informal.
- Shows how a tech hub can lead but still needs household reach.

RECOMMENDATIONS

- Set up e-kiosks and kerbside pickup for households.
- Require corporate take-back reporting.
- Register informal dealers as aggregators.

CONCLUSION

- Bengaluru shows a tech city can lead on formal recycling but needs household-level reach.

REFERENCES

- KSPCB - list of authorised e-waste recyclers
- CPCB - state-wise e-waste inventory
- E-Parisaraa Pvt Ltd - company material

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- IT hub means high corporate e-waste.
- KSPCB authorises; E-Parisaraa is the example.
- Gap: household e-waste goes to informal market.
- Fixes: e-kiosks, CSR, aggregators.
- Likely question: why is Bengaluru called the e-waste capital?

6 Mumbai's E-Waste Ecosystem

EXECUTIVE SUMMARY

- Mumbai combines high e-waste generation with a hybrid ecosystem of informal markets, refurbishers and authorised recyclers.

BACKGROUND

- Maharashtra ranks at or near the top of Indian states by e-waste volume (CPCB).
- Dense electronics markets (e.g., Lamington Road) and port-linked trade in used goods.
- Informal dismantling clusters in areas like Govandi, Mankhurd and Dharavi.

STRATEGY / INITIATIVE

- Authorised recyclers such as Eco Recycling Ltd (Ecoreco) process under MPCB authorisation.
- Repair and refurbishing markets extend product life.
- BMC and NGOs run collection drives and awareness events.

IMPACT & RESULTS

Indicator	Data / Outcome
State rank	Maharashtra among top e-waste generators (CPCB)
Informal clusters	Govandi, Mankhurd, Dharavi dismantling hubs
Formal player	Eco Recycling Ltd (Ecoreco), MPCB-authorised
Reuse channel	Repair and refurbishing markets (Lamington Road)

SWOT ANALYSIS

Strengths: Strong repair-and-reuse culture, many recyclers, port and market access

Weaknesses: Hazardous informal dismantling; weak household channels

Opportunities: Merge informal workers into formal chains; urban mining

Threats: Acid leaching, open burning, illegal imports

COMPARISON WITH INDIA

- Similar to Delhi-NCR and Bengaluru but with a denser informal cluster.
- Unlike Zurich, formal take-back is not the default route.
- Reuse culture is a strength India can build on.

RECOMMENDATIONS

- Register informal dismantlers as authorised collectors.
- Create ward-level drop-off points.
- Provide safety training and protective equipment.

CONCLUSION

- Mumbai's hybrid system shows both potential and risk.
- Integrating informal workers safely is the key step.

REFERENCES

- MPCB - authorised e-waste recyclers
- CPCB - e-waste inventory
- Eco Recycling Ltd (Ecoreco) - company reports

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- Hybrid: informal + refurbishers + authorised recyclers.
- Maharashtra is a top generator.
- Hazards: acid leaching, open burning.
- Fix: register informal workers, ward drop-offs.
- Likely question: describe Mumbai's e-waste chain.

7 Kerala's Clean Kerala Company

EXECUTIVE SUMMARY

- Clean Kerala Company Ltd (CKCL), set up by the state government, builds a state-wide channel for non-biodegradable waste and e-waste.
- It works with Haritha Karma Sena (HKS) collectors and local bodies.

BACKGROUND

- Formed in 2007 under the Local Self-Government Department as a government company.
- High literacy, dense settlement and limited landfill space.
- Works alongside the Suchitwa Mission and Haritha Keralam Mission.

STRATEGY / INITIATIVE

- HKS (mostly women) collect non-biodegradable waste door-to-door for a user fee.
- Material collection facilities and CKCL move waste to authorised recyclers.
- Periodic e-waste drives with local bodies and authorised recyclers.

IMPACT & RESULTS

Indicator	Data / Outcome
Set up	2007, Government of Kerala company
Collectors	HKS: ~30,000+ members, mostly women (approx.)
Coverage	Most local bodies in Kerala (approx.)
Revenue model	User fees + sale of recyclables

SWOT ANALYSIS

Strengths: Government backing, community workforce, high public awareness

Weaknesses: Uneven fee compliance; e-waste small next to plastics

Opportunities: Link with EPR producers; app-based pickup

Threats: Funding gaps, falling prices of recyclables

COMPARISON WITH INDIA

- State-led community model; other states depend more on private recyclers.
- Closer to Pune's SWaCH than Bengaluru's corporate model.
- Shows community collectors can scale statewide.

RECOMMENDATIONS

- Replicate HKS-style collectors in other states.
- Add e-waste as a fixed periodic pickup category.
- Link with EPR producers for funding.

CONCLUSION

- CKCL shows how a state agency plus community workers can run reliable collection.
- E-waste needs a stronger, EPR-funded stream.

REFERENCES

- Clean Kerala Company Ltd - official website
- Suchitwa Mission, Government of Kerala
- Kerala LSGD - Haritha Keralam Mission

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- CKCL: 2007, government company.
- Haritha Karma Sena = door-to-door collectors.
- Funding: user fees + recyclables.
- Strength: state + community partnership.
- Likely question: how does CKCL organise waste collection?

8 Pune's SWaCH Model

EXECUTIVE SUMMARY

- SWaCH is India's first wholly owned cooperative of self-employed waste pickers, partnered with Pune Municipal Corporation (PMC).
- It gives waste pickers formal identity and income while delivering door-to-door service.

BACKGROUND

- Waste picker union KKPKP began organising in the 1990s.
- SWaCH cooperative was formed in 2008 in partnership with PMC.
- Focus on segregated collection and dignity of work.

STRATEGY / INITIATIVE

- Members collect segregated waste door-to-door and charge households a monthly fee.
- PMC recognises SWaCH through an agreement and ID cards.
- Dry waste, including small e-waste, is routed to recyclers.

IMPACT & RESULTS

Indicator	Data / Outcome
Model	Cooperative of self-employed waste pickers (2008)
Members	~3,000+ waste pickers (approx.)
Reach	Roughly a million households in Pune (approx.)
Income	User fees + scrap sale

SWOT ANALYSIS

Strengths: Worker ownership, city partnership, strong civil society backing

Weaknesses: Dependence on PMC contracts; fee collection challenges

Opportunities: Add e-waste collection; replicate in other cities

Threats: Privatisation of city contracts, policy shifts

COMPARISON WITH INDIA

- A model of informal-sector integration; most Indian cities still exclude waste pickers.
- Contrasts with Switzerland's fully formal system.
- Shows inclusion and efficiency can go together.

RECOMMENDATIONS

- Replicate the cooperative model in other cities.
- Give waste pickers formal e-waste collection roles under EPR.
- Secure long-term municipal contracts.

CONCLUSION

- SWaCH proves waste pickers can be partners, not problems.
- Scaling it needs municipal commitment and EPR funding.

REFERENCES

- SWaCH Seva Sahakari Sanstha - official material
- KKPKP - waste picker union documents
- Pune Municipal Corporation - solid waste management

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- SWaCH = cooperative of waste pickers with PMC (2008).
- Income: user fee + scrap sale.
- Benefits: ID, dignity, segregation.
- Threat: privatisation.
- Likely question: why is SWaCH a model for India?

9 Dharavi Recycling Ecosystem

EXECUTIVE SUMMARY

- Dharavi is one of Asia's largest informal recycling clusters, handling plastics, metals, paper and some e-waste.
- It shows the economic power of the informal sector and the risks it faces.

BACKGROUND

- Dense area in central Mumbai, roughly 2 sq km (approx.).
- Thousands of small units sort, clean, shred and resell materials.
- Redevelopment plans threaten its business clusters.

STRATEGY / INITIATIVE

- Decentralised chain: waste pickers, sorters, small processors, traders, manufacturers.
- Low-cost, labour-intensive, near-zero landfill.
- For e-waste: manual dismantling, metal recovery and part resale.

IMPACT & RESULTS

Indicator	Data / Outcome
Businesses	Estimated ~5,000-15,000 small units (estimates vary)
Workers	Estimated 250,000+ in linked jobs (estimates vary)
Turnover	Often cited ~US$1 billion/year (estimate)
Recycling share	Large share of Mumbai's plastic waste (60%+ estimated)

SWOT ANALYSIS

Strengths: Low cost, large workforce, quick reuse, near-zero waste chain

Weaknesses: Unsafe conditions, pollution, no legal recognition

Opportunities: Formalise via cooperatives and EPR partners; safety training

Threats: Redevelopment displacement, hazardous e-waste practices, health risks

COMPARISON WITH INDIA

- Informal model recovers value cheaply but with health and environmental cost.
- Formal models (Japan, Switzerland) are safer but costlier.
- India needs a hybrid: keep informal reach, add formal standards.

RECOMMENDATIONS

- Recognise and register units as EPR collectors or aggregators.
- Set up safe dismantling zones with protective equipment.
- Protect livelihoods in redevelopment plans.

CONCLUSION

- Dharavi shows informal recycling is efficient but unsafe.
- Formalising it, not replacing it, is the sustainable path.

REFERENCES

- Municipal Corporation of Greater Mumbai - reports
- Academic studies on Dharavi's recycling economy
- CPCB - e-waste guidelines

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- Dharavi = large informal recycling cluster.
- Chain: pickers to sorters to processors to traders.
- Pros: low cost, jobs. Cons: health, pollution.
- Fix: hybrid formal-informal model.
- Likely question: strengths and risks of Dharavi.

10 Dell's Circular Economy Strategy

EXECUTIVE SUMMARY

- Dell combines take-back, recycled materials and design for reuse to move toward a circular economy.
- It links business goals with e-waste reduction.

BACKGROUND

- Dell has offered take-back since the 1990s; Dell Reconnect with Goodwill began in 2004.
- 2030 goals: 100% recycled or renewable packaging and >50% recycled or renewable product content.
- Global scale gives it influence on suppliers.

STRATEGY / INITIATIVE

- Free consumer recycling and mail-back; business trade-in and asset recovery.
- Closed-loop materials: recycled plastics, aluminium, carbon fibre, ocean-bound plastic.
- Responsible-recycling partners with data destruction.

IMPACT & RESULTS

Indicator	Data / Outcome
Lifetime recycling	Dell reports >2 billion lb of used electronics recovered
2030 packaging goal	100% recycled or renewable
2030 product goal	>50% recycled or renewable content
Approach	Take-back + recycled content + design for reuse

SWOT ANALYSIS

Strengths: Global scale, closed-loop supply chain, strong brand trust

Weaknesses: Recovery depends on customer participation; self-reported targets

Opportunities: Refurbished sales, device-as-a-service, EPR leadership in India

Threats: Rising material costs, greenwashing risk, competition

COMPARISON WITH INDIA

- Most Indian producers meet EPR via certificates; Dell-style closed loops go beyond compliance.
- Dell India offers take-back through authorised recyclers under EPR.
- Shows producers can profit from circular design.

RECOMMENDATIONS

- Encourage producers to publish verified recycled-content data.
- Promote trade-in and refurbished models in India.
- Design for repair and modular upgrade.

CONCLUSION

- Dell shows producer-led circularity is feasible at scale.
- Independent verification is needed for credibility.

REFERENCES

- Dell Technologies - ESG and circular economy reports
- Ellen MacArthur Foundation - circular economy
- CPCB - EPR guidelines

Figures are approximate; verify against the latest sources before submitting.

ONE-PAGE EXAM REVISION SHEET

- Circular economy = keep materials in use.
- Dell tools: take-back, recycled content, modular design.
- 2030 goals: 100% packaging, >50% product content.
- Link to EPR: beyond compliance.
- Likely question: evaluate Dell's circular strategy.